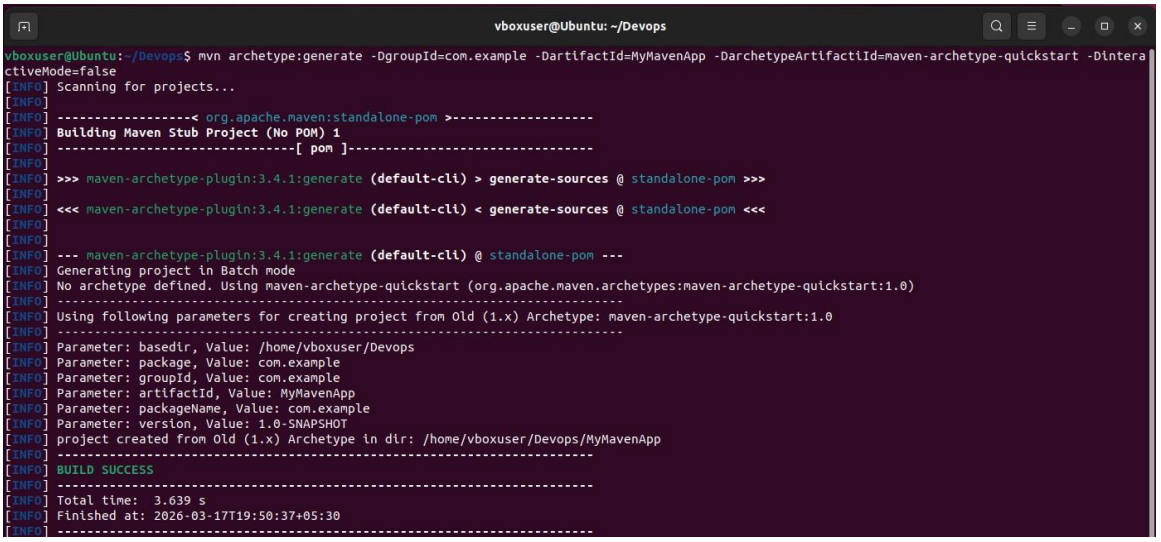
Project 1:Working with Maven: Creating a Maven Project, Understanding the POM File

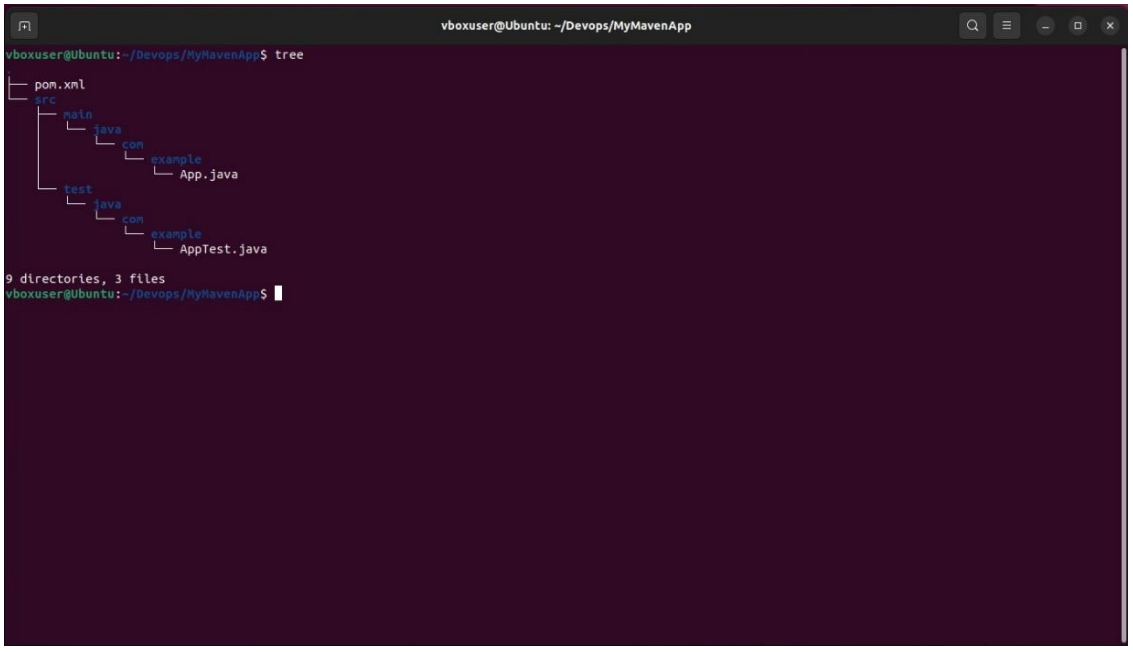
Dependency Management and Plugins

Step 1: Create a Maven project by running following command into terminal:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false



Running this command sets up a standard Maven project layout that includes a sample App.java and AppTest.java files.



A "BUILD SUCCESS" message in the terminal confirms the project was set up without any issues.

Step 2: Go to the project directory using command cd MyMavenApp

Step 3: Open and edit pom. xml file using command gedit pom.xml

- Go to https://mvnrepository.com
- Search for a dependency “junit”
- Select the latest version and copy the dependency XML snippet.
- Paste it under the <dependencies> section in pom.xml
- Updated pom.xml:

Open pom.xml ~/DevOps/MyMavenApp Save

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>


  <name>MyMavenApp</name>
  <url>http://maven.apache.org</url>

  <!-- Properties: Customize Java version or plugin versions -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <!-- Build: Configuring plugins and build settings -->
<build>
  <plugins>

    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.8.1</version>
    </plugin>

    <plugin>
      <groupId>org.codehaus.mojo</groupId>
      <artifactId>exec-maven-plugin</artifactId>
      <version>3.1.0</version>
      <configuration>
        <mainClass>com.example.MainClass</mainClass>
      </configuration>
    </plugin>
```

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Update App.java:

Open App.java ~/Desktop/Devops/Lab1/MyMavenApp/src/main/java/com/example

```java
package com.example;

/**
 * Hello world!
 *
 */
public class App
{
    public static void main( String[] args )
    {
        System.out.println( "Hello World! This is the content of MainClass.java" );
    }
}
```

Step 4: Build the project using command mvn compile

Maven will begin compiling the source files, and the output will appear as shown below.

```
vboxuser@Ubuntu: ~/Desktop/Devops/Lab1/MyMavenApp
vboxuser@Ubuntu:~/Desktop/Devops/Lab1/MyMavenApp$ mvn compile
[INFO] Scanning for projects...
[WARNING]
[WARNING] Some problems were encountered while building the effective model for com.example:MyMavenApp:jar:1.0-SNAPSHOT
[WARNING] 'dependencies.dependency.(groupId:artifactId:type:classifier)' must be unique: junit:junit:jar -> version 4.13.2 vs 3.8.1 @ line 18, column 17
[WARNING]
[WARNING] It is highly recommended to fix these problems because they threaten the stability of your build.
[WARNING]
[WARNING] For this reason, future Maven versions might no longer support building such malformed projects.
[WARNING]
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/Desktop/Devops/Lab1/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.749 s
[INFO] Finished at: 2026-03-17T19:54:10+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@Ubuntu:~/Desktop/Devops/Lab1/MyMavenApp$
```

Once compilation finishes, a "BUILD SUCCESS" message appears, confirming the build completed without errors.

Step 5: Package the project using command mvn package.

Step 6: Run the test on project using command mvn test.

Step 7: To run the Maven application

- Go to target directory present in MyMavenApp using command cd target
- Execute the JAR file with the command:

  java -jar MyMavenApp-1.0-SNAPSHOT.jar
- Once execution is complete, the terminal will display the result based on the code in App.java.

The output depends on the logic inside App.java. You can modify the file contents to get a different result each time.

```
vboxuser@Ubuntu:~/Desktop/Devops/Lab1/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello World! This is the content of MainClass.java
vboxuser@Ubuntu:~/Desktop/Devops/Lab1/MyMavenApp$
```